Project 3: Exploring the Generative Power of Diffusion Models: From Latent Space to Multimodal Synthesis

Prof. Yuijao Shi

October 18, 2025

Instructions:

1. Deadline: 2025-11-16 11:59 pm
2. You need to write your report in LaTeX. Word is acceptable, but the final submission must be a PDF. All text must be in English.
3. Your report should be submitted in PDF format and packed with your code. The naming format of the compressed file should be studentID-name-project3.zip.
4. Please submit your zip file to link https://epan.shanghaitech.edu.cn/l/W11wly

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1 Introduction: Generative Models on MNIST

This project delves into the realm of generative models, focusing on the powerful capabilities of Diffusion Models. The entire assignment is anchored to the MNIST dataset, serving as a clean, illustrative testbed for validating the core concepts and advanced applications of generation.

The **Basic** section will first introduce the Variational Autoencoder (VAE) [4], exploring its fundamental abilities in image reconstruction and latent space compression. Following this, we will implement and compare the widely adopted Denoising Diffusion Probabilistic Models (DDPM) [3] and its accelerated counterpart, Denoising Diffusion Implicit Models (DDIM) [7]. Ultimately, we will integrate VAE and DDPM to construct a more efficient latent space diffusion pipeline.

In the **Advanced** section, we transition to multimodal generation by training a simplified text-to-image model, `tiny-mnist-sd`, which conceptually aligns with the architecture of models like Stable-Diffusion-3 [2] but is drastically simplified for MNIST.

Furthermore, we will critically compare the architectural designs of several cutting-edge text-to-image models and explore the practical application of the ComfyUI inference framework [1]. A comparative analysis between ComfyUI's workflow and our `tiny-mnist-sd` pipeline will be conducted.

Note: The assignment includes a compressed file containing the foundational code. Please ensure you thoroughly read the accompanying `README` *file.*

2 Basic Explorations: Foundations of Generation

2.1 VAE: Reconstruction and Latent Encoding

This section utilizes the provided `train_vae.py` script. A pre-implemented VAE model and its training script are given. After setting up the environment(following the `README` file!), minor parameter adjustments are encouraged. The VAE loss function is primarily composed of two terms: the **Reconstruction Loss** and the **KL Divergence Loss**.

- **Report Task 1:** Clearly explain the mathematical meaning and intuition behind both the Reconstruction Loss and the KL Divergence Loss within the context of VAEs.
- **Report Task 2:** Experiment with adjusting the β parameter in the code, which governs the weighting of the KL Divergence term. Compare and contrast the resulting image reconstruction quality and the quality of samples generated from the latent space across different β values. Illustrate the trade-off between reconstruction fidelity and latent space regularity.

2.2 DDPM and DDIM: Accelerating Generative Synthesis

This task is based on the `train_diffusion.py` script. DDPM represents the foundational sampling strategy for diffusion models but suffers from low efficiency due to the necessity of numerous sequential sampling steps. The provided code implements the complete training and DDPM sampling procedure.

- **Implementation Task:** Extend the existing DDPM framework by implementing the DDIM sampler. DDIM significantly improves sampling efficiency by allowing for non-Markovian generation steps, skipping intermediate steps.
- **Report Task:** Demonstrate and compare the sampling results of your implemented DDIM against the baseline DDPM. While DDIM results may be slightly inferior, ensure that the generated images retain clear recognizable features (e.g., more than 1/3 of the samples are discernible). Discuss the theoretical advantage of DDIM regarding computational efficiency.

2.3 Latent Space Diffusion: Enhancing Efficiency

This section involves the `train_latent_diffusion.py` script. For high-resolution image generation (e.g., 1080P), generating noise and sampling directly in the pixel space is computationally demanding and complicates network learning due to low information density. A widely adopted alternative is Latent Space Diffusion, which leverages an Encoder-Decoder pair to compress the pixel space into a smaller, information-dense latent space. Diffusion is then performed in this efficient latent space.

The provided code uses the VAE trained in Section 2.1 as the frozen Encoder and Decoder.

- **Implementation Task:** Based on your experiments in Section 2.1, select and train an appropriate VAE model for this latent diffusion stage.
- **Hint:** When a VAE is repurposed as a latent space compressor for diffusion, its role closely mimics that of a standard Autoencoder (AE). Focus on maximizing its **reconstruction capability** by adjusting the VAE parameters, potentially favoring a smaller β to prioritize the reconstruction term.
- **Report Task:** Justify your chosen VAE parameters and demonstrate the quality of the generated samples from the latent diffusion pipeline.

3 Advanced Applications: Controllable Generation

3.1 Conditional Generation: Training a Tiny Multimodal SD

The basic framework achieved random sample generation. To enable user-defined output, this section introduces conditional control, culminating in the training of a simplified text-to-image model, `tiny-mnist-sd`. This requires incorporating a CLIP-like [5] conditioning mechanism into the UNet [6].

- **Prerequisites:** Run `train_clip.py` to obtain a CLIP model trained on MNIST images and their corresponding labels. No modifications to this file are needed.
- **Implementation Task:** Complete the missing code in `Unet.py` by implementing the logic for injecting the CLIP condition into the UNet structure. The overall pipeline is set up in `train_tiny_mnist_sd.py`.
- **Validation and Report:** After successful training, the model should demonstrate controlled generation, typically producing a grid of 10 columns (one for each digit 0-9) for a given batch. Validate that the generated digits align with the conditioning labels, aiming for at least 1/3 of the samples to be correctly recognizable and controlled.

3.2 ComfyUI: A Glimpse into Diffusion Workflows

ComfyUI is a highly flexible, node-based inference framework for Stable Diffusion, allowing users to construct complex generative workflows by chaining various components (VAEs, checkpoints, LoRAs, samplers, etc.). A typical workflow might involve: generating a rough latent image using positive/negative CLIP prompts, refining the latent with LoRA style models, and finally decoding with a high-resolution upscaler.

- **Practical Task:** Install ComfyUI (https://github.com/comfyanonymous/ComfyUI). Construct **at least two different** text-to-image workflow and generate some images. You may need to analyze the failure cases and try some possible solutions (e.g., attaching different LoRA models)
- **Report Task:** Compare and contrast the workflow structure in ComfyUI with the pipeline of our implemented `tiny-mnist-sd`. Discuss the conceptual similarities (e.g., separate conditioning, UNet, VAE stages) and the key differences (e.g., flexibility, component swapping, explicit control flows).

References

[1] comfyanonymous. Comfyui. 2023.

[2] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*, 2024.

[3] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.

[4] Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.

[5] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.

[6] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015.

[7] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.